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# Feasibility of infrared spectroscopy for discrimination between gallbladder polyp and gallbladder stone using bile juices

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ABSTRACT

This study has attempted infrared (IR) spectroscopic identification of gallbladder (GB) polyp and GB stone using bile juices as specimens. For drop-n-dry measurement of bile juice samples, Si wafer patterned with circular (3 mm in diameter) spots surrounded by hydrophobic coating was prepared to make the formation of sample droplets consistent in shape, and IR spectra of the dried samples on the substrate were acquired. Several noticeable differences in IR spectral features between GB polyp and GB stone samples were observed. First, the intensities of amide I and II bands considerably differed, thereby indicating the differences in the protein composition in both samples. Second, the absorption of $COO^-$ stretching bands arising from amino acid-conjugated bile salts, able to form micelles and increase their solubility, was higher for GB polyp samples, equivalently indicating the less chance of GB stone. Third, the intensities of phospholipids bands were smaller in the GB stone samples since the lower phospholipid concentration led to the decrease of cholesterol solubilization and thus increased the chance of GB stone. The first five principal component scores that describe 95.3% of the total spectral variance were fed into support vector machine (SVM) as a classifier. The cross-validated discrimination accuracy was 78.6%. To improve the accuracy further, a sample pretreatment step capable of enhancing the compositional dissimilarity is necessary for some samples that show similar spectral features.

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## 1. Introduction

Bile juice is an abundant body fluid and continuously produced by the liver. It is stored in the gallbladder and excreted into the intestine through a common bile duct into the duodenum [1–7]. It has two major functions. First, bile salts in bile juice help digestion since these are essential cofactors for pancreatic lipase. Bile salts are amphipathic, detergent-like molecules that make lipids soluble by forming micelles [8,9]. Second, bile juice has an excretion function for various compounds from endogenous (e.g., cholesterol, steroid hormones and bilirubin) or exogenous (e.g., drugs and environmental chemicals) origin, either as native substances or liver-derived metabolites [10,11].

Bile juice consists of ~85% water and contains many important substances for its function. The dissolved substances are composed of 61% bile salts (forms of bile acids conjugated with amino acids), 12% fatty acids, 9% cholesterol, 7% proteins, 3% phospholipids (mainly phosphatidylcholine), 3% bilirubin and others, and their composition varies according to the nutritional states of individuals [7]. Cholic and chenodeoxycholic acids are the primary human bile acids. Deoxycholic and lithocholic acids are secondary bile acids [12]. These bile acids are conjugated with taurine or glycine in the liver, thereby forming bile salts. Both cholesterol and phosphatidylcholine form small unilamellar vesicles (40–200 nm in diameter) in bile juice. Bile salts secreted from hepatocytes convert these vesicles into smaller micelle structures (40–100 Å in diameter) due to their detergent-like property. Thus, various micelles and the remaining unilamellar vesicles are present in bile juice as cholesterol carriers [2,13–18].

Since the composition of bile juice could be closely related with gallbladder pathological conditions, bile juice is a valuable specimen for relevant disease diagnosis and relatively easy to obtain. Nevertheless, only a few studies on bile juice based on mass spectrometry (MS) have been reported [19–24]. In the earlier stage, MS was used to identify different bile acids [5,25,26] and study biliary bile acids in the bile juice of gallstone patients before and during treatment with chenodeoxycholic acid [27]. Recently, MS-based proteomic studies were performed to (a) identify various proteins in gallbladder bile juice samples of cholesterol gallstone patients [19], (b) investigate conspicuous overexpression of CEACAM6 (carcinoembryonic antigen-related cell adhesion molecule 6) and MUC1 (mucin 1) CA 19-9 (carbohydrate antigen 19-9) in pancreatic cancer and cholangiocarcinoma (bile duct cancer) samples [22], and (c) clarify the differentially expressed proteins (human serum albumin, profilin and retinol binding protein) between the cholesterol gallstone and control groups [23]. In these studies, hyphenated separation methods of sodium dodecyl sulfate-polyacrylamide gel electrophoresis (SDS-PAGE), two-

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dimensional gel electrophoresis (2DE) and liquid chromatography (LC) were also necessary. Consequently, overall analytical systems were not only costly but they also required extensive sample pretreatment steps. Although MS is surely a potential method, analytical methods capable of providing diagnostic information with much simpler measurement would be further beneficial.

Infrared (IR) spectroscopy is a simple and fast method with no or minimal sample preparation for measurement of medical samples including bile juices. Nonetheless, these methods have rarely been studied for diagnosis of various hepatopancreatico-biliary (HPB) diseases. Quite recently, only one research on IR spectroscopic differentiation between benign and malignant bile duct strictures has been reported [28]. They analyzed raw bile juice samples as well as chloroform-extracted samples, and found that the absorbance related to phospholipids was less intense in malignant samples compared to that in benign samples.

In this paper, we have conducted a feasibility study on IR spectroscopy-based differentiation of gallbladder (GB) polyp and GB stone using bile juices potentially as a simple and effective method in diagnosis of HPB diseases. Since the major component in GB bile juices is water, broad and intense water absorption limits the available spectral window for IR spectroscopic measurement and eventually degrades the signal-to-noise ratio of spectra. Therefore, a drop-n-dry measurement based on spectral acquisition of dried samples on a substrate after solvent (water) evaporation was adopted to eliminate the above-mentioned problems. For the measurement, a circular spot (3 mm in diameter) patterned Si wafer covered with a hydrophobic coating except the spot was prepared. A sample droplet was confined within the circular spot due to the surrounding hydrophobic environment, and the droplet formation was more consistent by using the substrate. The bile juices obtained from GB polyp and GB stone patients were dried on the substrate, and then the corresponding IR spectra were collected using a microscope. The sharp peaks originated from ambient water were substantially eliminated in prior using wavelet transformation (WT) for clear comparison of spectral features [29,30]. The spectral differences between GB polyp and GB stone samples were initially searched, and the distribution of principal component scores obtained from all IR spectra by principal component analysis (PCA) was also examined. Then, support vector machine (SVM) was used to estimate the accuracy of identification of both diseases.

## 2. Experimental

### 2.1. Sample collection

All of the samples used in this study were acquired from the Hanyang University Hospital (Seoul, Korea). The sample set composed of bile juices was obtained from 9 GB polyp patients (5 males and 4 females, ages from 24 to 68 years) and 27 GB stone patients (12 males and 15 females, ages from 22 to 85 years). In the case of GB stone, 10, 11 and 6 had cholesterol gallstones, pigment gallstones, and both of them, respectively. Each sample was acquired from the gallbladder after cholecystectomies, and immediately frozen and stored at −80 °C for subsequent spectroscopic measurement. The informed consent form was signed by every patient enrolled in this study. The clinical data of the patients and samples including serum and bile juices were collected prospectively after the permission from institutional review board of Hanyang University. Fig. 1 shows the pictures of all bile juice samples contained in glass vials. The sample colors are mostly brown, greenish yellow and dark green, and there is no visual distinction between GB polyp and GB stone samples.

### 2.2. Preparation of patterned Si-wafer and spectral acquisition of IR spectra

Circular patterns (3 mm in diameter) on a Si wafer with a thin layer of the SU-8 50 photoresist (thermally stable epoxy based negative UV-photoresist, Microchem, Newton, MA, USA) were prepared according to the manufacturer's instruction. A Si wafer was treated with oxygen plasma for 75 s and then transferred to the spinner for coating of SU-8 50 photoresist. The thickness of the coating was approximately 100 μm. Photolithography was performed on the photoresist-coated Si wafer using a UV mask aligner (MDA-400, MIDAS System, Korea) with a piece of film photomask combined with a glass substrate. It was exposed to UV with the i-line (365 nm) through the film mask for 30 s, and baked on a hot plate at 65 °C for 2 min, and 95 °C for 15 min. The photoresist was developed in propylene glycol methyl ether acetate (PGMEA) over 1–2 min at room temperature with mild agitation and rinsed with isopropyl alcohol.

After thawing the samples, a 1 μl of bile juice was dropped on a circle pattern on the substrate, and the sample droplet was allowed to dry in an ambient condition over 30 min. Spectra of the dried sample on the substrate were collected for analysis. IR spectral acquisition was performed on a transmission mode using a VERTEX 70 FTIR spectrometer equipped with Hyperion 2000 FTIR microscope (Bruker Optics Korea). Sample spectra were collected by accumulating 32 scans with a resolution of 4 $cm^{-1}$. All spectral pretreatments (WT, baseline correction, and area normalization), principal component analysis (PCA), and support vector machine (SVM) were performed using MATLAB R2012b (MathWorks Inc., MA, USA).

## 3. Results and discussion

### 3.1. IR measurements of bile juice samples and examination of spectral features

For drop-n-dry measurement, one GB stone sample was dropped on the circle-patterned Si wafer and dried for IR spectral collection. The images of the original sample in a glass vial and dried sample on the substrate are shown in Fig. 2(a). The dried sample spot is clearly circular due to confinement of the raw sample within the spot. When a sample was allowed to dry on the substrate, the distribution of components inside the spot was inhomogeneous due to the dissimilar interactions of the components with the surface during sample drying. As shown in the image, the outer dark circle is resulted from the thicker aggregation of the components in the course of water evaporation. In addition, the distribution of dried sample color is not homogeneous within the spot.

To examine the possible compositional inhomogeneity, seven transmission IR spectra (1800–890 $cm^{-1}$ range) were collected along the diameter line of the spot using a microscope with intervals of 300 μm as shown in Fig. 2(a). The aperture size was also 300 μm. The spectral collection points are designated on the image by the dots with different colors, and the corresponding spectra are indicated with the same colors. No useful spectra were acquired at the outer dark circle due to the thick sample condensation that results in the saturation of absorption bands. It is observed that the intensities (absorbances) of spectra vary, depending on the spectral collection points. This result informs that the thickness of the dried sample is not uniform over the spot.

To compensate the intensity variation by the different sample thicknesses, the baselines of spectra were initially corrected at 1800 and 890 $cm^{-1}$ and normalized by the area under the 1800–890 $cm^{-1}$ range as shown in Fig. 2(b). After the normalization, the overall spectral features are similar to one another although there are still some differences in the absorption magnitude, which indicates that the compositional inhomogeneity within the spot is not significant while the variation of sample thickness occurs as a result of drying.

Fig. 3(a) shows the IR spectra collected at 7 different points on a dried sample of another GB stone sample. The images of the corresponding raw sample and its dried form in the spot are also shown. In comparison with the sample presented in Fig. 2, it was less viscous and its color was brighter. The color of the dried spot is bright light

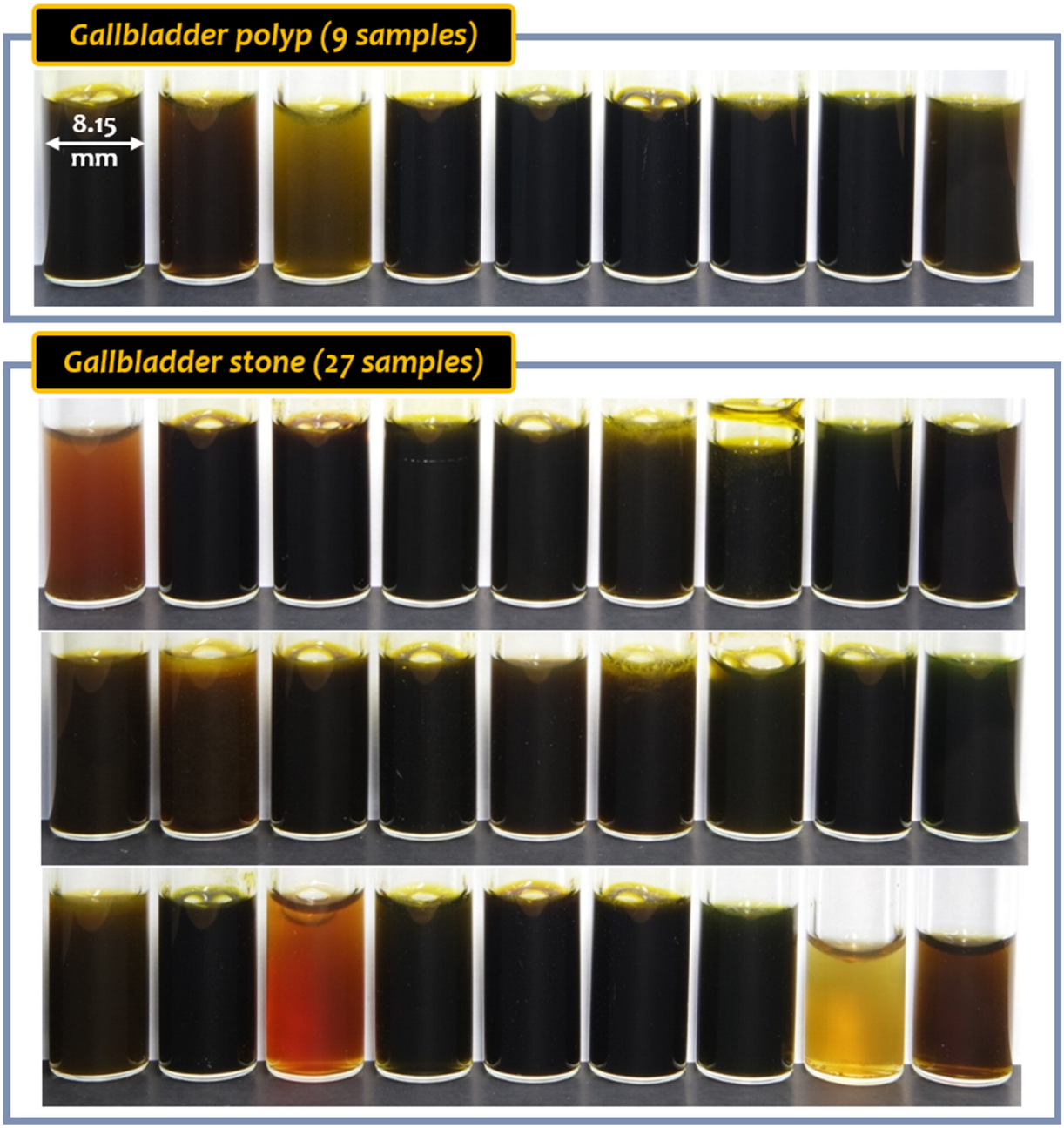


**Fig. 1.** The pictures of all the bile juice samples contained in glass vials (diameter: 8.15 mm). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

yellow, thus indicating a less dense sample coverage on the surface. Overall absorbances of spectra are lower compared to those in Fig. 2(a) due to the thinner sample coverage (thickness). The absorbances of spectra acquired around the middle (pink) are relatively lower while those around the edge (black/red) are higher. It means that the sample thickness decreases from the edge as it reaches the center in this case. Therefore, the noise becomes more apparent over the 1800–1300 $cm^{-1}$ range in the spectra collected around the center. The corresponding normalized spectra (bottom) are also shown in Fig. 3(b).

The noise-like sharp peaks are resulted from the absorption of ambient water and necessary to eliminate or minimize, since they would negatively impact on accuracy of the discrimination. In this study, wavelet transformation (WT) able to effectively diminish sharp noise-like feature present in spectra was employed. It decomposes an original spectrum using a wavelet, i.e., the projection onto a wavelet basis function. Then, out of all wavelet coefficients, specific wavelet coefficients associated with such as noise are discarded in a course of inverse transformation back to the original domain. By this way, selective removal of specific components such as de-noising could be achievable. Discrete wavelet transformation (DWT) using Daubechies wavelet (order: 7, level: 8) was chosen as previously adopted for the same purpose [30].

In the WT, three-level detail coefficients ($d^1$, $d^2$, $d^3$) that were presumed to be original spectral noise and sharp ambient water peaks were removed. The spectra (top) in Fig. 3(b) are the WT-treated spectra of the original spectra (bottom). The WT-treated spectra are offset for comparison with clarity. As clearly observed that the sharp water peaks are nearly disappeared and the bands in the 1800–1300 $cm^{-1}$ range becomes more apparent without distortion in overall spectral

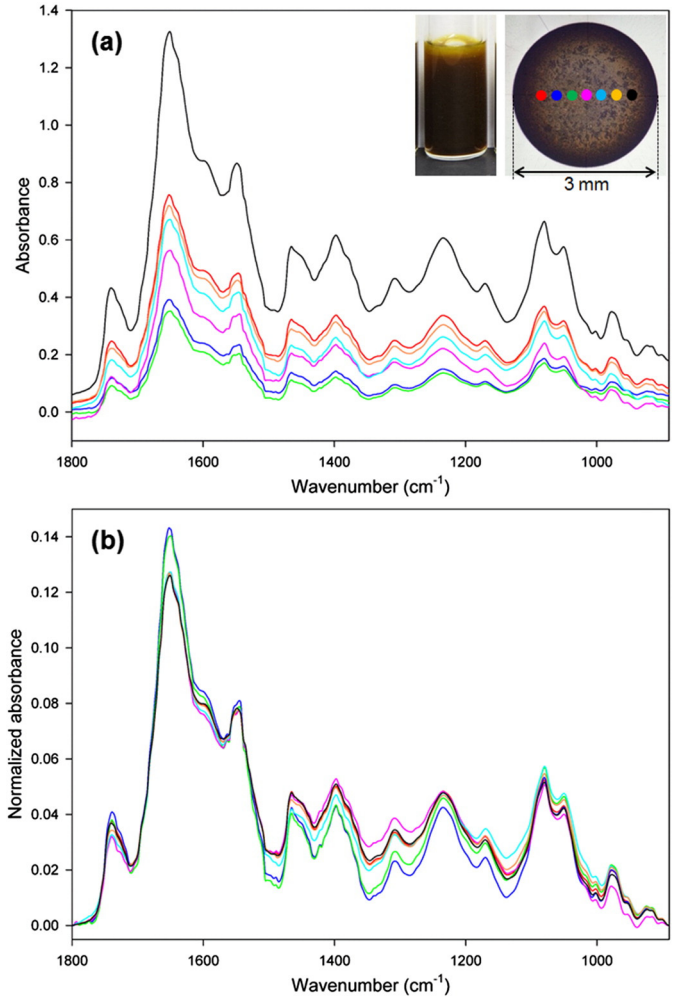


**Fig. 2.** Transmission IR spectra collected from a dried GB stone sample (a). The sample viscosity is relatively high. The inset images show the raw sample in a glass vial (left) and the drop-dried sample on the substrate (right). The dots with different colors indicate the locations where IR spectra are collected, and the corresponding spectra are designated using the same colors. The normalized spectra are also shown (b). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

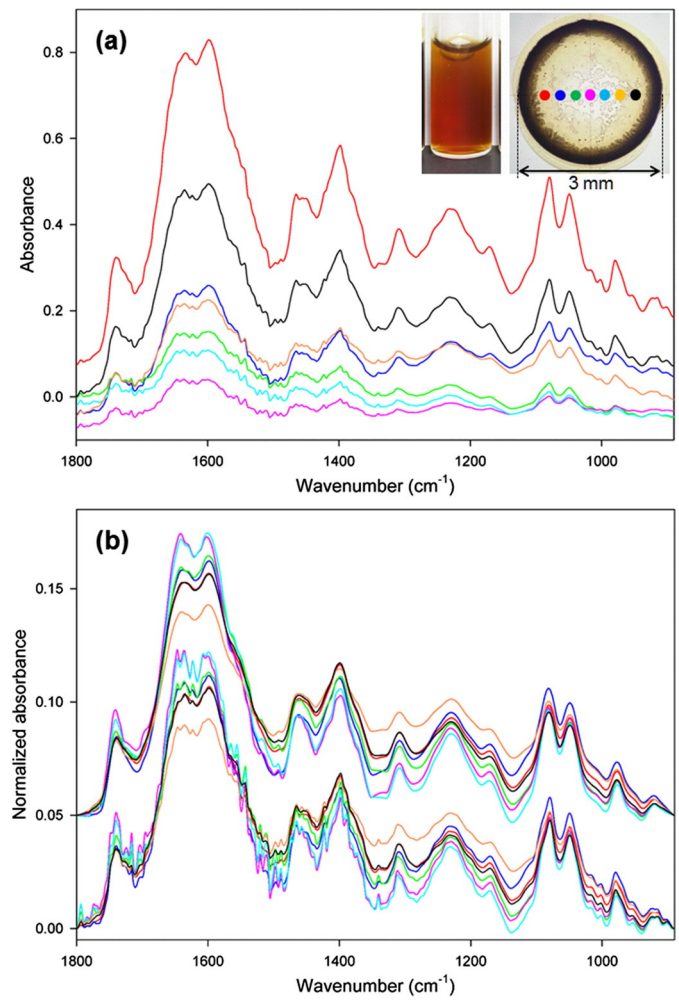


**Fig. 3.** Transmission IR spectra collected from a dried GB stone sample (a). The sample viscosity is relatively low. The inset images show the raw sample in a glass vial (left) and the drop-dried sample on the substrate (right). The dots with different colors indicate the locations where IR spectra are collected, and the corresponding spectra are designated using the same colors. The normalized spectra (bottom) and the corresponding WT-treated spectra (top) are also shown (b). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

feature after WT. The spectral features of the spectra are similar to each other except the differences in their absorption magnitude. It confirms that the compositional inhomogeneity over the spot is insignificant. The effective suppression of the water peaks by WT provides a clearer vision in differentiating the spectral features between GB polyp and GB stone samples. For further analysis, WT-treated spectra were solely used and, for each sample, three spectra with higher absorption magnitude were selected out of the 7 spectra for the dataset construction.

The colors of all 36 samples used in this study are mainly brown, greenish yellow, dark green as shown in Fig. 1. It seems that the brown samples are mixed with a small amount of blood due to inflammation. Approximately one third of the samples had relatively lower viscosity as in the sample in Fig. 3. Fig. 4 shows average spectra (WT-treated spectra) of all the GB polyp (red) and GB stone samples (blue). The shade with the same color indicates the corresponding standard deviation. All normalized spectra were used for the averaging. The average spectrum of the GB stone samples is offset for comparison. The same average spectra without the shaded standard deviations and the baseline offset are shown in Fig. 4(b) for an easier comparison of both spectra.

The spectral features of the GB polyp and GB stone samples are fairly different over the spectral range. The amide I band (1649 and 1635 $cm^{-1}$) corresponding to the C═O stretching vibrations of the peptide bonds modulated by the secondary structures (α-helix, β-sheet, etc.) and the amide II band (1542 $cm^{-1}$) arising from the C—N stretching vibrations in combination with N-H bending [31–33] are clearly present. The amide I band is more dominant in the average spectrum of the GB stone samples, while the amide II band is relatively stronger for the GB polyp samples. This observation informs that the protein compositions in the GB polyp and GB stone samples are dissimilar to each other, and the spectral difference is a valuable source for discrimination.

In a meanwhile, cholate and chenodeoxycholate are primary bile salts, consisting of 80% of the total bile salts [34,35]. They can be converted to glycocholate and glycochenodeoxycholate by glycine conjugation, and their anti-symmetric and symmetric $COO^-$ stretching bands are observed at 1601 $cm^{-1}$ and 1400 $cm^{-1}$, respectively. Such amino acid-conjugated bile salts are located at lipid/water interface and able to form micelles at a proper condition as reported earlier. The formation of micelles decreases the chance for GB stone due to the increased solubility. Therefore, the higher intensities of these bands in the case of GB polyp, corresponding to the larger amount of amino acid-conjugated bile salts, are observed as shown in Fig. 4(b).

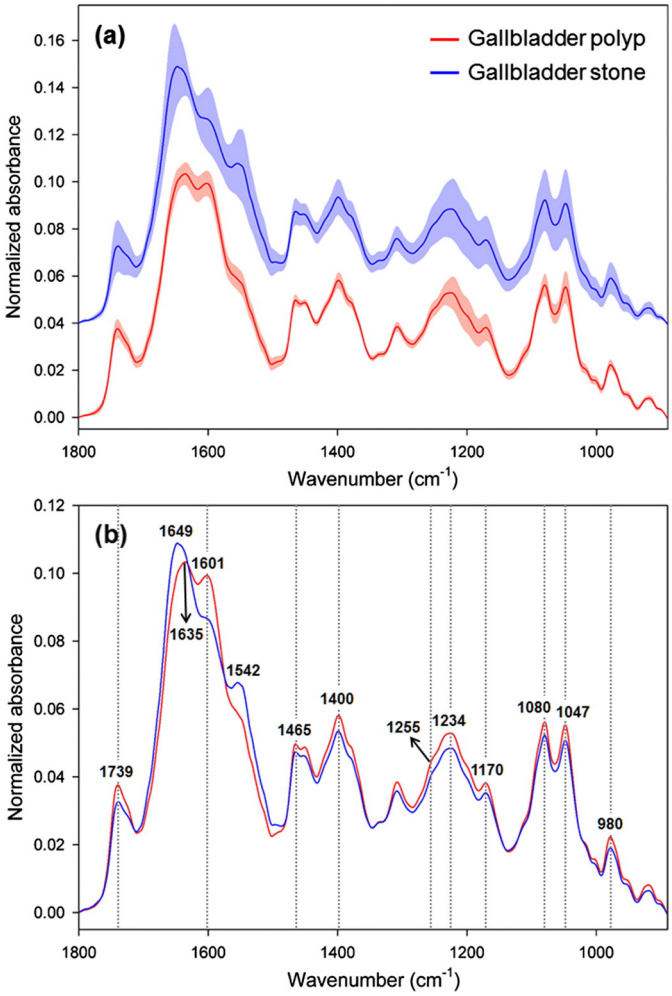


**Fig. 4.** Average IR spectra of all GB polyp (red) and GB stone samples (blue, offset for easier comparison). The shade with the same color indicates the corresponding standard deviation (a). The same average spectra without the shaded standard deviations and the baseline offset are also shown (b). The locations of major bands are also indicated. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

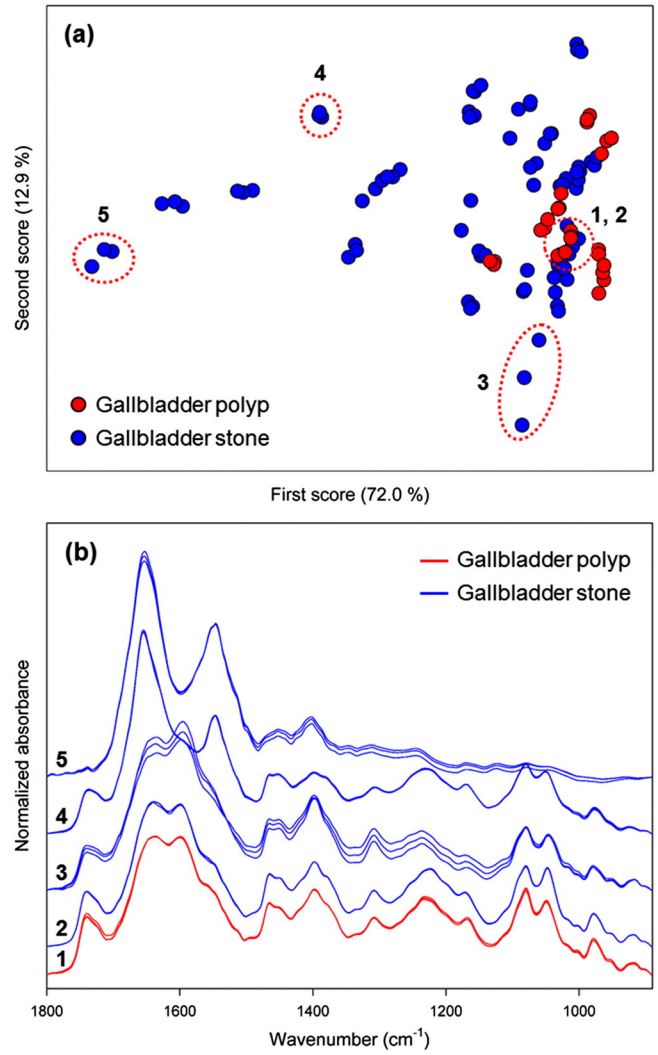


**Fig. 5.** Two-dimensional scatter plot built by using the 1st and 2nd scores obtained from all sample spectra using PCA (a). Red and blue scores indicate GB polyp and GB stone samples, respectively. IR spectra of the marked samples by dotted lines on the scatter plot are shown (b). The spectra are arbitrarily offset for comparison. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

In addition, phospholipid related bands are present over the spectral ranges such as the peaks at 1739, 1465, 1255, 1234, 1170, 1080, 1047 and 980 $cm^{-1}$ [36]. The peaks at 1739 $cm^{-1}$ and 1465 $cm^{-1}$ are from the carbonyl C═O stretching vibrations of ester functional groups in lipids [28] and C—H bending vibration, respectively. The 1255/1234 $cm^{-1}$ and 1170 $cm^{-1}$ peaks correspond to the P═O vibration and $PO_2^-$ vibrations, respectively. The peaks at 1080, 1047, 980 $cm^{-1}$ are originated from the P—O—C asymmetric and symmetric stretching bands of phosphate group [36].

Wang et al. used the model bile juice system to investigate the pathway of cholesterol crystallization (eventually leading to macroscopic gallstones) and construct the ternary phase equilibrium diagram among bile salts, phospholipids, and cholesterol [37,38]. They reported that the relative amount of bile salts and phospholipids was related with the solubilization of cholesterol, for example, the larger concentration of phospholipids relative to that of bile salts enhanced the cholesterol solubilization, hence less chance of GB stone [2,13]. The intensities of phospholipids bands are marginally lower for the GB stone samples as shown in Fig. 4(b). This result is in accordance with the above report that the decreased phospholipid concentration lowers the degree of cholesterol solubilization, and thus the higher chance of cholesterol crystallization leading to GB stone is consequential.

### *3.2. Examination of accuracy for discrimination between GB polyps and GB stones*

To easily and simultaneously investigate the spectral dissimilarities of all samples, the principal component analysis (PCA) was performed for the dataset, and the subsequent scores were used for the examination. Fig. 5(a) shows two-dimensional scatter plot built by using the 1st and 2nd scores. The numbers in the parentheses indicate the percentages of spectral variations described by the corresponding factors relative to the total spectral variations in the dataset. Red and blue circles correspond to the GB polyp and GB stone samples, respectively. As expected, the score distribution of GB stone samples is much greater compared to that of the GB polyp samples due to the larger spectral variations among the samples as shown in Fig. 4(a); while the scores of both groups around the right middle are overlapped with each other.

For detailed examination, five samples (three samples from around the edge of the GB stone score group, two samples from the both score-overlapped areas) are selected as marked by dotted lines, and the corresponding spectra are shown in Fig. 5(b). In the cases of samples

4 and 5, the spectral features are considerably different from those of other three samples, especially in the 1800–1350 $cm^{-1}$ range. The intensities of 1649 and 1542 $cm^{-1}$ bands are strong. The spectral feature of sample 3 is fairly different from those of samples 4 and 5, and it becomes relatively closer to those of samples 1 and 2. The band shapes of samples 1 and 2 are nearly similar as expected. It is worthwhile to indicate that the samples 3, 4 and 5 were obtained from the subjects of pigment gallstones. Therefore, it informs that the compositions of pigment as well as cholesterol gallstone samples are considerably dissimilar in each case due to different pathological pathways [39,40]. In overall, the examination of the figure reveals that the discrimination between GB polyp and GB stone is feasible by directly measuring bile juice samples using IR spectroscopy without further sample pretreatment although some samples with similar spectral features still remain difficult to distinguish.

To classify the samples into either GB polyp or GB stone, SVM was employed in this study. SVM is a popular and powerful supervised classifier, which finds the best hyper-plane to classify samples by maximizing the distance between class boundaries [41,42]. The first five principal component scores that describe 95.3% of the total spectral variance in the dataset were fed into SVM. The discrimination accuracy was obtained by three-fold cross-validation. A calibration set included 66.7% of the total samples (6 samples out of the 9 GB polyp samples in total, 19 samples out of the 28 GB stone samples in total), and the remaining samples were used for validation. The discrimination accuracy obtained by the cross-validation was 78.6%. It is important to note that, out of all 9 GB polyp patients, 4 patients had both GB polyps and GB stones; therefore, the presence of these samples could make the discrimination more difficult and negatively influence on the accuracy. In overall, the discrimination of GB polyp or GB stone has become feasible although its accuracy needs to be improved further. It is clear that samples with closely located scores such as samples 1 and 2 also lower the accuracy. Therefore, it is necessary to develop an additional sample pretreatment step that is capable of enhancing the compositional dissimilarity of these samples for more accurate diagnosis of GB polyp and GB stone, such as removal of macromolecules in a bile juice to highlight identity of small molecules (metabolites).

## 4. Conclusions

In this study, we demonstrated the feasibility of IR spectroscopy for discrimination between GB polyp and GB stone by using raw bile juice samples. The differences in the protein composition, relative amount of amino acid-conjugated bile salts and relative concentration of phospholipids were the main sources that enabled the discrimination. On the other hand, some GB polyp and GB stone samples showed similar spectral features and their discrimination was difficult. To resolve this problem, a sample pretreatment step capable of enhancing compositional dissimilarity between the samples is necessary, and it is now under investigation. Furthermore, to gain a greater statistical significance for the proposed IR spectroscopic method as a diagnosing tool, it is necessary to include a larger number of samples with greater compositional variability in the dataset. Detailed compositional analysis of each bile juice sample is also necessary. The dataset expansion with detailed compositional analysis of samples is underway as a continuing research to drive the proposed method further into a clinically acceptable stage for diagnosis of HPB diseases.

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